

When Perception Outperforms Action: Dissociating Rhythm Perception from Motor Synchronization

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INTRODUCTION

Humans perceive and spontaneously synchronize with rhythm, anticipating beat timing before it occurs, showing a predictive feature rare in other species. Prevalent theories propose that such rhythm perception depends on motor simulation, the ability to move with accurate timing [1]. However, clinical evidence shows individuals perceiving rhythm accurately despite impaired motor synchronization [2], suggesting a possible dissociation between perception and action. This raises a central question: Does rhythm perception require the ability to synchronize, and does synchronization actually reflect perceptual accuracy?

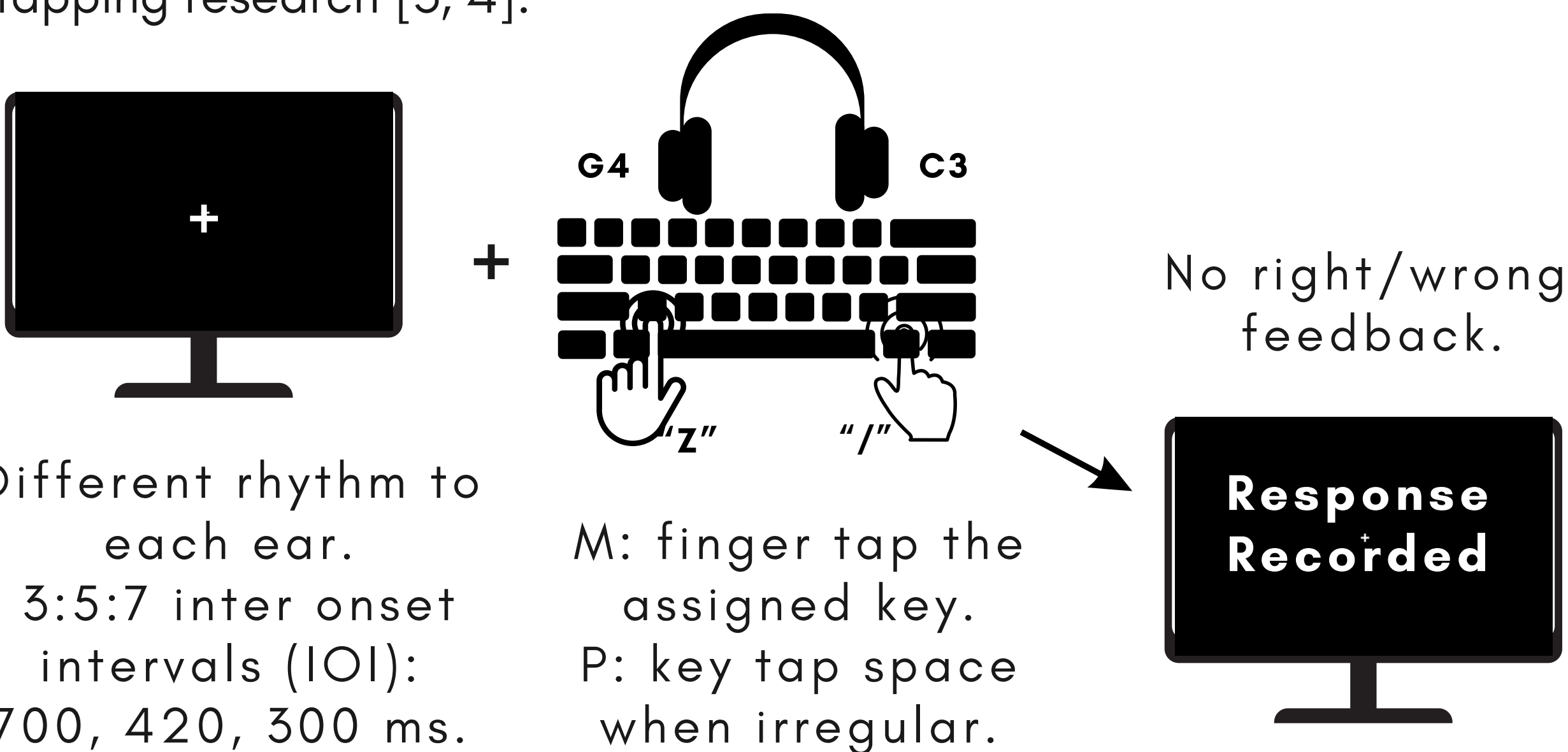
METHODOLOGY

Participants: N = 30 (F = 15), within-subjects design. No participants are musicians or have received formal training. Amateur musical experiences varied.

Stimuli: Dual-stream polyrhythms (3:5, 3:7, 5:7) presented dichotically. One stream to one ear, distinguished by pitch (C3 / G4). Streams counterbalanced across ears and pitch assignments.

Perception task: Detect 200 ms timing delays in either stream (72 trials: 48 perturbed, 24 control). Measured: hit rate, d' .

Motor task: Tap simultaneously to both streams, one pattern per hand (120 trials). Measured: success rate (IOI accuracy $\pm 25\%$, CV < 0.50, omission/extraneous rate < 40%). These thresholds were set deliberately generous to accommodate the difficulty of dual-stream polyrhythmic tapping, following tolerance windows established in prior polyrhythm tapping research [3, 4].



SUMMARY

Test whether accurate polyrhythm perception requires successful motor synchronization, using identical stimuli in both tasks within the same participants.

ANALYSIS

Two separate mixed-effects logistic regression models were fit respectively for perception and motor task. Each model included a random intercept per participant to account for individual baseline differences.

Perception accuracy ~

polyrhythm pairing + frequency assignment + which cycle was perturbed + which stream was perturbed + which ear was perturbed + musical training + (random intercept | participant)

Motor tapping success ~

polyrhythm ratio + hand-to-ear mapping + rhythm role (fast/slow stream) $\times$ pitch + musical training + (random intercept | participant)

CONCLUSION

Accurate rhythm perception does not require successful motor synchronization. The two abilities are statistically and behaviorally dissociable.

Implication: Perceptual polyrhythm representations are robust with high irregularity detection accuracy; motor execution appears to hit a bottleneck that does not align with the performance level of perception. This challenges the assumption that synchronization performance reliably indexes perceptual ability.

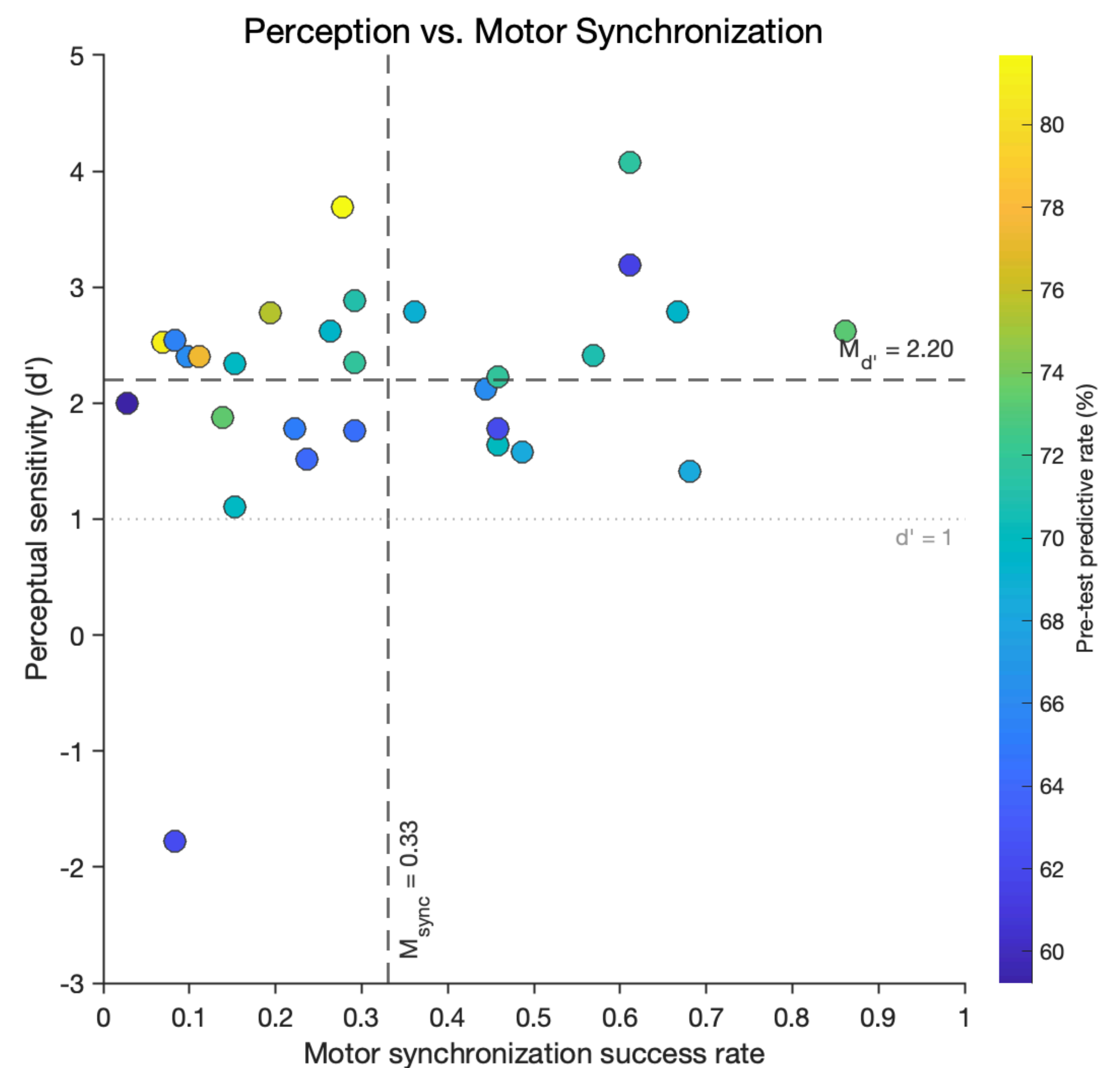
Future direction : Whether this dissociation reflects constraints in motor execution (down stream) or higher-level planning (upper stream) remains an open question for future work.

RESULTS

Perceptual-motor dissociation: Nearly all participants showed **high** perceptual sensitivity (mean $d' = 2.2$, success rate = 0.77), while motor synchronization was highly variable and frequently **poor** (mean success rate = 0.33). The same individuals who perceived well often failed to synchronize.

Training dissociation: Musical training did **not** predict perceptual accuracy ($p = .33$) but significantly predicted motor success ($p = .001$). All above-average synchronizers had musical experience, but musical experience did not guarantee synchronization success.

Stimulus effects: Perception: fast-stream perturbations were easier to detect ($F = 114$, $p < .001$), and left-ear perturbations were more salient than right ($F = 12.6$, $p < .001$). Motor: tapping the fast stream was easier than the slow ($F = 83.6$, $p < .001$), and contralateral hand-to-ear mapping was harder than ipsilateral ($F = 41.2$, $p < .001$). There was a significant interaction between rhythm and pitch identity ($F = 24.7$, $p < .001$), meaning that when tapping to the slower stream (longer IOI interval), the low-pitched stream's demand was smaller than expected.



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